

Two-Stage Autoignition and Negative Temperature Coefficient (NTC) Behaviour of Dimethyl Ether (DME)/Air Mixtures

A Homogeneous-Reactor Kinetic Study in Cantera

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Abstract

The low-temperature autoignition of dimethyl ether (DME, CH_3OCH_3) in air is studied numerically with the open-source chemical-kinetics library CANTERA. A zero-dimensional, adiabatic, constant-pressure homogeneous reactor is used to compute ignition delay times (IDT) over the temperature range 625–1300 K, at pressures of 10, 20 and 40 bar and equivalence ratios of 0.5, 1.0 and 2.0, with the hierarchically validated NUI Galway 2015 reaction mechanism (1268 species, 5336 reactions), which contains the complete low-temperature DME sub-model. The simulations reproduce the characteristic *negative temperature coefficient* (NTC) region, in which the ignition delay paradoxically increases with temperature, and the associated two-stage (cool-flame) ignition. The strength of the NTC behaviour weakens and shifts to higher temperature as pressure rises, while the ignition delay decreases monotonically with both pressure and equivalence ratio. The cool-flame chemistry is traced to the build-up and decomposition of the $\text{CH}_3\text{OCH}_2\text{O}_2$ peroxy radical and of formaldehyde.

1 Introduction

Dimethyl ether is the simplest ether and an attractive alternative fuel: it can be produced from biomass, natural gas or captured CO_2 , it has a high cetane number (≈ 55 –60), and its soot-free combustion makes it a candidate for compression-ignition (diesel) and homogeneous-charge compression-ignition (HCCI) engines. Because DME ignites through the same low-temperature radical chemistry as large alkanes, it is also a key benchmark fuel for validating detailed kinetic mechanisms.

A defining feature of such fuels is the **negative temperature coefficient (NTC)** of the ignition delay: over an intermediate temperature window (roughly 750–1000 K) the overall reactivity decreases as the temperature rises, so that the mixture ignites *more slowly* when it is hotter. The NTC region is accompanied by **two-stage ignition**, in which a weak, low-temperature heat release — the *cool flame* — precedes the main ignition event. These phenomena directly control engine knock, HCCI combustion phasing and the operability limits of advanced low-temperature combustion engines, which is why their accurate prediction is of practical importance.

The aim of this project is to reproduce and analyse the NTC behaviour and the two-stage autoignition of DME/air mixtures using a detailed chemical-kinetics model, and to quantify how the ignition delay and the strength of the NTC region depend on temperature, pressure and equivalence ratio. All computations are carried out with CANTERA in Python; the full source code accompanies this report.

2 Literature review

The low-temperature oxidation of DME was characterised experimentally and kinetically by Curran et al. [1], who established the degenerate chain-branching pathway responsible for cool flames. The comprehensive mechanism of Zhao et al. [2] (55 species, 290 reactions) became the standard reference model for DME and reproduces shock-tube ignition delays, flow-reactor speciation and laminar flame speeds. Burke et al. [3] extended the validation to high pressures (up to 110 bar) for methane, DME and their blends, and their NUI Galway model forms the basis of the AramcoMech family of hierarchically constructed mechanisms [9]. Reduced models suitable for CFD, such as the 30-species skeletal mechanism of Bhagatwala et al. [5] and the reduced model of Pan et al. [4], retain the low-temperature chemistry required to capture the NTC region.

The chemical origin of the NTC behaviour is now well understood [6, 7]. At low temperature the fuel radical $R^\bullet$ (here $\text{CH}_3\text{OCH}_2^\bullet$) adds molecular oxygen to form a peroxy radical $\text{RO}_2^\bullet$, which isomerises to a hydroperoxyalkyl radical $^\bullet\text{QOOH}$ and, after a second O_2 addition, leads to chain branching through ketohydroperoxide decomposition. As the temperature rises, the $\text{RO}_2 \rightleftharpoons R + \text{O}_2$ equilibrium shifts back towards the reactants, suppressing this branching pathway; the system must then wait for the slower intermediate-temperature H_2O_2 chemistry to take over, which is the microscopic cause of the increase in ignition delay with temperature.

3 Numerical model

3.1 Software and chemistry

All computations use CANTERA 3.2 with the NUI Galway 2015 hierarchical mechanism (the *n*-hexane-NUIG-2015 model distributed with Cantera as example data), comprising 1268 species and 5336 elementary reactions and built on the comprehensively validated AramcoMech C_0 – C_2 core [9]. Although developed for the hexane isomers, the model contains the complete low-temperature DME sub-mechanism (the $\text{CH}_3\text{OCH}_2\text{O}_2$ / $\text{CH}_3\text{OCH}_2\text{O}_2\text{H}$ / ketohydroperoxide pathway); DME (CH_3OCH_3) is therefore taken directly as the fuel. Air is modelled as $\text{O}_2:\text{N}_2 = 1:3.76$.

3.2 Reactor model and governing equations

Autoignition is modelled with a homogeneous (zero-dimensional), adiabatic, constant-pressure batch reactor, which represents an isolated fluid element of a premixed charge. For a closed adiabatic system at constant pressure the governing equations solved by Cantera are the species and energy balances

$$\frac{dY_k}{dt} = \frac{\dot{\omega}_k W_k}{\rho}, \quad c_p \frac{dT}{dt} = -\frac{1}{\rho} \sum_k h_k \dot{\omega}_k W_k, \quad (1)$$

where Y_k , W_k , h_k and $\dot{\omega}_k$ are the mass fraction, molar mass, specific enthalpy and net molar production rate of species k , ρ is the density, c_p the mixture specific heat and T the temperature. The resulting stiff system of ODEs is integrated with the CVODES solver (absolute tolerance 10^{-15} , relative tolerance 10^{-6}).

3.3 Ignition-delay definition and two-stage detection

The total ignition delay time (IDT) is defined as the instant of maximum temperature-rise rate, i.e. the time at which dT/dt reaches its global maximum. The first-stage (cool-flame) delay is identified as the first significant local maximum of dT/dt preceding the main event; a cool flame is reported only when this early peak is clearly separated from the main ignition. This dual criterion extracts both stages of a two-stage ignition automatically from a single reactor

integration. Initial temperatures were varied from 625 to 1300 K (28 points); the pressure was set to 10, 20 and 40 bar at stoichiometric conditions, and the equivalence ratio to $\phi = 0.5, 1.0, 2.0$ at 20 bar — in total 5 temperature sweeps (≈ 140 reactor simulations) plus three time-resolved ignition histories.

4 Results and discussion

4.1 NTC region and the effects of pressure and equivalence ratio

Figure 1 shows the computed ignition delay of DME/air as a function of inverse temperature. All curves display the classic S-shape: a high-temperature branch (left), the NTC region in the middle, where the delay increases with temperature, and a low-temperature branch (right). The NTC region is most pronounced at the lowest pressure (Fig. 1a) and persists at every equivalence ratio (Fig. 1b).

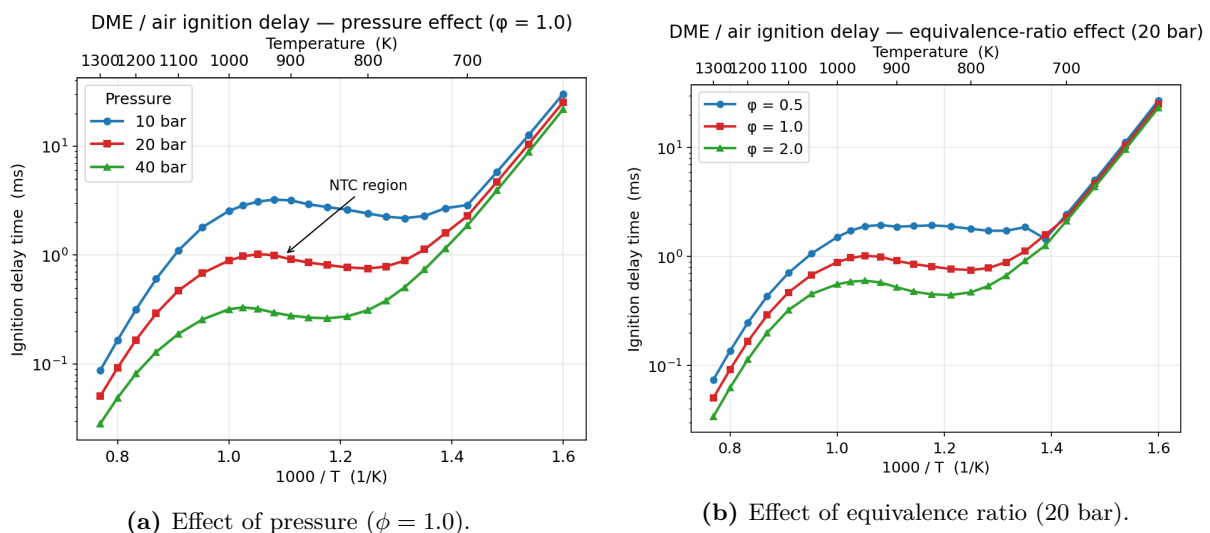


Figure 1: Ignition delay of DME/air versus inverse temperature. The non-monotonic NTC region is visible in every case.

Table 1 quantifies the NTC region. As pressure rises from 10 to 40 bar, the whole curve shifts to shorter delays (higher reactivity), the NTC zone moves to higher temperature (valley from 760 to 850 K), and the strength of the NTC — the ratio of the local-maximum to the local-minimum delay — weakens from 1.48 to 1.26. This is the expected behaviour: higher pressure favours the chain-branching O_2 addition steps and broadens the temperature window over which they remain active. Increasing the equivalence ratio shortens the ignition delay throughout the high-temperature and NTC regions — at 1000 K the delay falls from 1.51 ms ($\phi = 0.5$) to 0.89 ms ($\phi = 1.0$) and 0.56 ms ($\phi = 2.0$) — because a higher fuel concentration accelerates the rate-limiting fuel + OH and RO_2 isomerisation steps. On the low-temperature branch the three ϕ curves converge, indicating that there the delay is governed by the intrinsic low-temperature chain chemistry rather than by the amount of fuel present.

Table 1: Location and strength of the NTC region (stoichiometric DME/air).

Pressure (bar)	Valley T (K)	Valley IDT (ms)	Peak T (K)	Peak/valley
10	760	2.18	925	1.48
20	800	0.75	950	1.35
40	850	0.26	975	1.26

4.2 Two-stage (cool-flame) ignition and low-temperature chemistry

Figure 2 presents the time history of a representative case inside the NTC window ($T_0 = 850$ K, 20 bar, $\phi = 1.0$). The temperature trace exhibits two distinct rises: a small step of about 100 K at ≈ 0.15 ms — the cool flame — followed by a thermal-runaway main ignition at ≈ 0.81 ms. The lower panel, showing dT/dt on a logarithmic scale, makes the two heat-release pulses unambiguous. The cool flame partially consumes the fuel and raises the temperature into the range where the low-temperature branching becomes inefficient, introducing the induction period that separates the two stages — the very mechanism that produces the NTC behaviour of Figure 1.

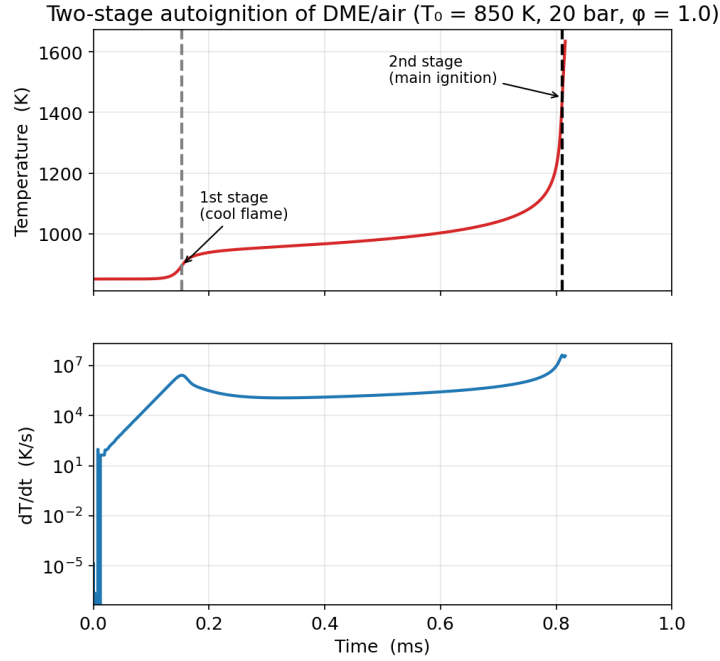


Figure 2: Two-stage autoignition of DME/air (850 K, 20 bar, $\phi = 1.0$). Top: temperature; bottom: temperature-rise rate. The cool flame precedes the main ignition by about 0.65 ms.

Figure 3a follows the key species during the same ignition. During the cool-flame stage the peroxy radical $\text{CH}_3\text{OCH}_2\text{O}_2$ (scaled $\times 50$) rises and falls in a sharp pulse, while formaldehyde CH_2O accumulates — both are the fingerprints of the RO_2/QOOH low-temperature pathway. Only a fraction of the DME is consumed at this stage. The accumulated intermediates (CH_2O , H_2O_2) then decompose, the OH ($\times 20$) and CO pools explode, the remaining fuel and O_2 are consumed and the temperature runs away to the adiabatic flame value — the main ignition. Figure 3b plots the cool-flame delay together with the total delay: a distinct first stage is detected only inside the NTC window ($\approx 720\text{--}900$ K); the cool-flame delay decreases monotonically with temperature, whereas the total delay turns over, confirming that it is the lengthening induction *between* the two stages, not the cool flame itself, that creates the NTC behaviour.

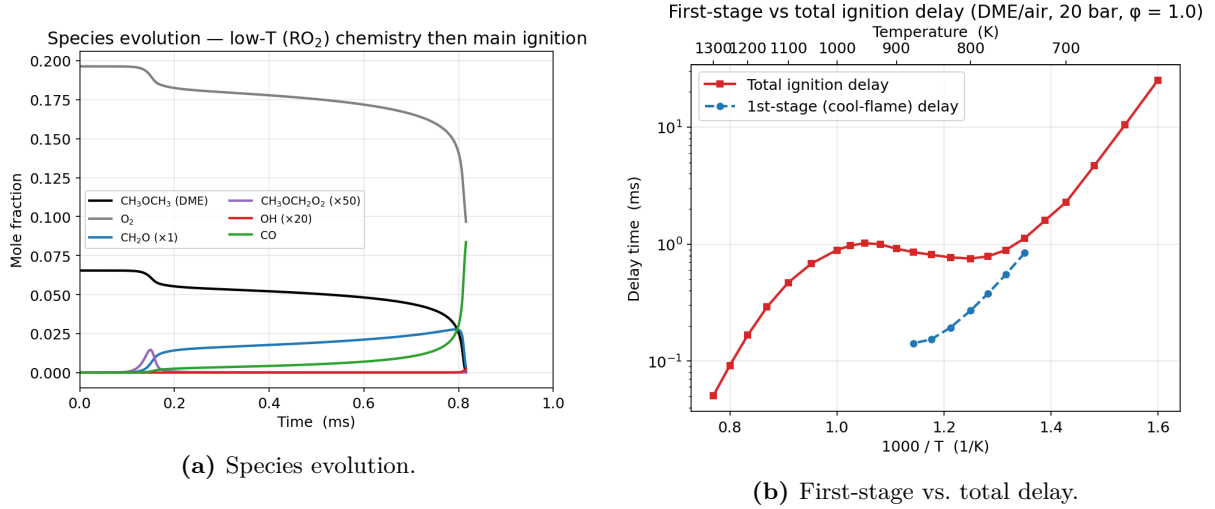


Figure 3: Low-temperature chemistry. Left: mole-fraction history at 850 K, 20 bar (the CH₃OCH₂O₂ pulse and CH₂O build-up mark the cool flame). Right: the cool flame appears only inside the NTC window.

4.3 Reaction-rate sensitivity analysis

To identify the reactions that control the NTC behaviour, a brute-force sensitivity analysis was performed at a representative point inside the NTC window (825 K, 20 bar, $\phi = 1.0$). Each rate coefficient k_i of the DME sub-mechanism and of the principal small-molecule branching reactions was multiplied by a factor of two and the normalised sensitivity $S_i = \partial \ln \tau / \partial \ln k_i$ of the total ignition delay τ was evaluated. The twelve most influential reactions are shown in Figure 4; a negative S_i means the reaction promotes ignition (shortens τ), a positive S_i that it inhibits it.

The result is exactly the textbook picture of low-temperature chain branching. Ignition is promoted most strongly by the second oxygen addition $\text{CH}_2\text{OCH}_2\text{O}_2\text{H} + \text{O}_2 \rightarrow \text{O}_2\text{CH}_2\text{OCH}_2\text{O}_2\text{H}$ ($S = -0.76$), by the fuel-consumption step $\text{CH}_3\text{OCH}_3 + \text{OH} \rightarrow \text{CH}_3\text{OCH}_2 + \text{H}_2\text{O}$ ($S = -0.53$), by the hydrogen-peroxide branching $\text{H}_2\text{O}_2(+\text{M}) \rightarrow 2\text{OH}$ ($S = -0.40$) and by the first O₂ addition and its isomerisation ($\text{CH}_3\text{OCH}_2 + \text{O}_2 \rightarrow \text{CH}_3\text{OCH}_2\text{O}_2 \rightarrow \text{CH}_2\text{OCH}_2\text{O}_2\text{H}$). It is inhibited by the chain-terminating decompositions that compete with these branching steps: the QOOH decomposition $\text{CH}_2\text{OCH}_2\text{O}_2\text{H} \rightarrow 2\text{CH}_2\text{O} + \text{OH}$ ($S = +0.71$), the fuel-radical β -scission $\text{CH}_3\text{OCH}_2 \rightarrow \text{CH}_2\text{O} + \text{CH}_3$ ($S = +0.43$) and the concerted elimination $\text{CH}_3\text{OCH}_2\text{O}_2 \rightarrow 2\text{CH}_2\text{O} + \text{OH}$ ($S = +0.33$). The NTC behaviour is the direct outcome of this competition: as the temperature rises, the chain-terminating β -scission and the reverse $\text{RO}_2 \rightarrow \text{R} + \text{O}_2$ step outpace the chain-branching O₂-addition route, so the ignition delay lengthens.

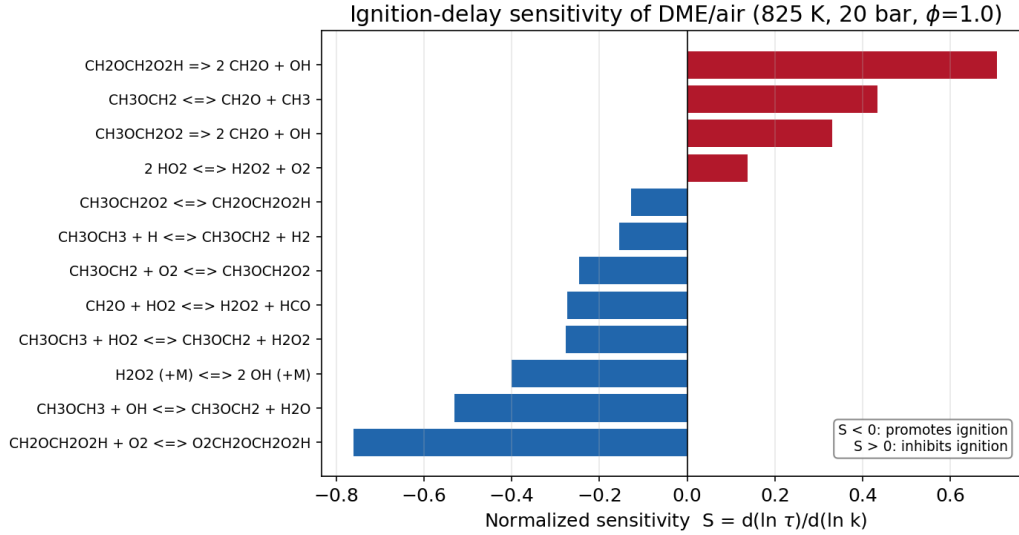


Figure 4: Brute-force sensitivity of the total ignition delay to individual reaction rates (825 K, 20 bar, $\phi = 1.0$; twelve most influential reactions). Blue (negative S): promotes ignition; red (positive S): inhibits ignition.

5 Validation against literature

Although the NUI Galway / AramcoMech mechanism used here has itself been validated extensively against shock-tube and rapid-compression-machine data [3, 9], two independent checks confirm that the present 0-D model reproduces the established DME behaviour (Figure 5). First, the apparent activation energy obtained from an Arrhenius fit of the high-temperature branch ($T \geq 1050$ K) is $E_a = 119$ kJ/mol, well within the 117–130 kJ/mol range reported from shock-tube measurements [11, 10]. Second, the computed NTC window (valley near 800 K, turnover near 950–975 K, shifting to higher temperature with pressure) coincides with the 800–1050 K NTC region measured by Pfahl et al. [10] for stoichiometric DME/air at 13 and 40 bar. The slightly lower turnover temperature obtained here is consistent with the use of a constant-pressure (rather than reflected-shock, constant-volume) reactor and of the maximum-gradient ignition criterion. The absolute delays — sub-millisecond at 20–40 bar and 1000–1200 K — are also of the magnitude measured in those experiments.

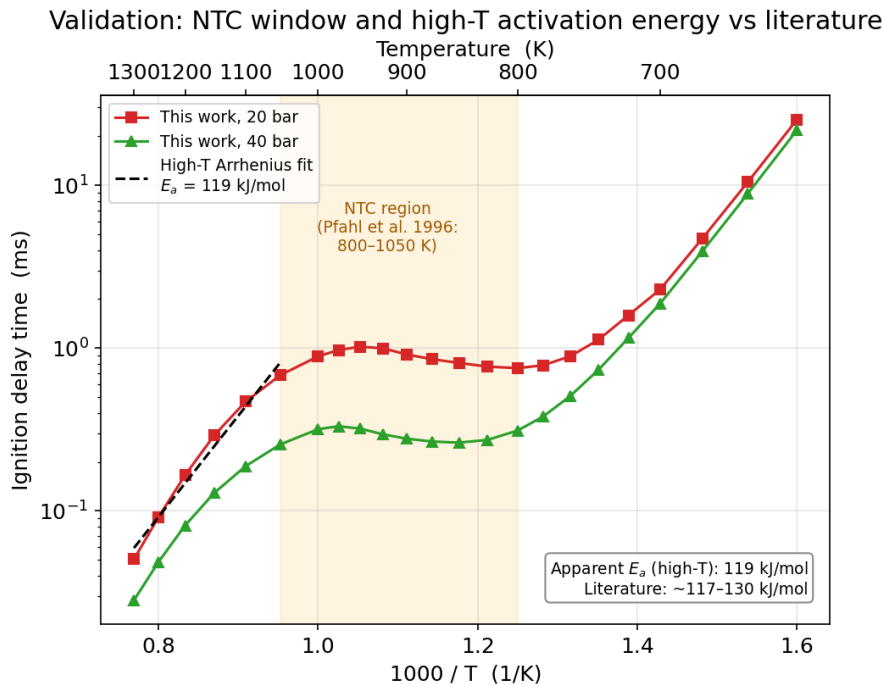


Figure 5: Validation. The computed high-temperature branch yields an apparent activation energy of 119 kJ/mol (literature 117–130 kJ/mol), and the predicted NTC region overlaps the 800–1050 K window measured by Pfahl et al. (1996).

6 Conclusions

A detailed chemical-kinetics study of DME/air autoignition was carried out in CANTERA using a zero-dimensional adiabatic constant-pressure reactor and the NUI Galway 2015 mechanism. The main findings are:

1. The model reproduces the characteristic NTC region of DME, in which the ignition delay increases with temperature between roughly 760 and 975 K.
2. Raising the pressure from 10 to 40 bar shortens the ignition delay, shifts the NTC region to higher temperature and weakens it (NTC strength falling from 1.48 to 1.26).
3. Richer mixtures ignite faster; the equivalence ratio has little effect on the low-temperature branch, where the intrinsic chain chemistry dominates.
4. Inside the NTC window the ignition is two-stage: a cool flame, driven by the $\text{CH}_3\text{OCH}_2\text{O}_2$ peroxy-radical chemistry and formaldehyde build-up, precedes the main ignition. The NTC behaviour arises from the lengthening induction period between the two stages as the RO_2 branching pathway is progressively shut off with rising temperature.
5. A brute-force sensitivity analysis identifies the controlling chemistry: the second O_2 addition and H_2O_2 branching promote ignition, while the QOOH decomposition and the CH_3OCH_2 β -scission inhibit it; their competition is the origin of the NTC. The model is consistent with the literature: an apparent high-temperature activation energy of 119 kJ/mol and an NTC window matching Pfahl et al. (1996).

The results are physically consistent and in qualitative agreement with the published DME literature [2, 3], demonstrating that 0-D homogeneous-reactor modelling in CANTERA captures the essential low-temperature combustion physics relevant to engine knock and HCCI combustion. The principal modelling limitation is the use of a detailed but generic NUIG mechanism rather

than a dedicated, separately optimised DME model; a constant-volume reactor and a sensitivity analysis of the controlling RO_2 reactions would further refine the absolute delay times without changing the qualitative conclusions.

References

- [1] H. J. Curran, S. L. Fischer, F. L. Dryer. The reaction kinetics of dimethyl ether. II: Low-temperature oxidation in flow reactors. *Int. J. Chem. Kinet.* 32 (2000) 741–759.
- [2] Z. Zhao, M. Chaos, A. Kazakov, F. L. Dryer. Thermal decomposition reaction and a comprehensive kinetic model of dimethyl ether. *Int. J. Chem. Kinet.* 40 (2008) 1–18.
- [3] U. Burke, K. P. Somers, P. O’Toole, et al. An ignition delay and kinetic modeling study of methane, dimethyl ether, and their mixtures at high pressures. *Combust. Flame* 162 (2015) 315–330.
- [4] L. Pan, S. Kokjohn, Z. Huang. Development and validation of a reduced chemical kinetic model for dimethyl ether combustion. *Fuel* 160 (2015) 165–177.
- [5] A. Bhagatwala, Z. Luo, H. Shen, J. A. Sutton, T. Lu, J. H. Chen. Numerical and experimental investigation of turbulent DME jet flames. *Proc. Combust. Inst.* 35 (2015) 1157–1166.
- [6] C. K. Westbrook. Chemical kinetics of hydrocarbon ignition in practical combustion systems. *Proc. Combust. Inst.* 28 (2000) 1563–1577.
- [7] J. F. Griffiths. Reduced kinetic models and their application to practical combustion systems. *Prog. Energy Combust. Sci.* 21 (1995) 25–107.
- [8] D. G. Goodwin, H. K. Moffat, I. Schoegl, R. L. Speth, B. W. Weber. *Cantera: An Object-oriented Software Toolkit for Chemical Kinetics, Thermodynamics, and Transport Processes*, version 3.2, 2024. <https://www.cantera.org>
- [9] W. K. Metcalfe, S. M. Burke, S. S. Ahmed, H. J. Curran. A hierarchical and comparative kinetic modeling study of C_1 – C_2 hydrocarbon and oxygenated fuels. *Int. J. Chem. Kinet.* 45 (2013) 638–675.
- [10] U. Pfahl, K. Fieweger, G. Adomeit. Self-ignition of diesel-relevant hydrocarbon-air mixtures under engine conditions. *Proc. Combust. Inst.* 26 (1996) 781–789.
- [11] R. D. Cook, D. F. Davidson, R. K. Hanson. Shock tube measurements of ignition delay times and OH time-histories in dimethyl ether oxidation. *Proc. Combust. Inst.* 32 (2009) 189–196.